

ABSTRACT

The **Casualty & Disaster Insurance Management System** is a full-stack enterprise application designed to automate and streamline the complete lifecycle of casualty and disaster insurance management. The system addresses critical challenges in traditional insurance workflows such as delayed claim processing, lack of disaster verification, fraud risks, and inefficient role-based coordination. Built using **.NET 8 Clean Architecture, SQL Server, and Angular**, the platform ensures scalability, security, and maintainability while providing a seamless digital experience for all stakeholders.

The system supports four specialized roles: **Admin, Agent, Claims Officer, and Customer**, each with clearly defined responsibilities and access controls enforced through JWT-based authentication and role-based authorization. Admin users manage policy products and oversee system analytics. Agents are responsible for selling policies, assisting customers, and tracking premium payments. Customers can purchase policies, submit claims, upload supporting documents, and track claim status in real time. Claims Officers review claims, analyze fraud risk scores, verify disaster authenticity, and make final approval decisions.

A key feature of the platform is its **Geographic Disaster Verification Engine**, which cross-checks the incident location and date of a submitted claim against real-time disaster event data obtained through external APIs. By utilizing stored address and geolocation coordinates, the system validates whether a reported disaster occurred at the specified location and time. This significantly reduces fraudulent claims and enhances trust in the insurance process.

The system also incorporates an intelligent **Fraud Detection Engine** that generates a risk score based on predefined rules and claim patterns. Suspicious claims are automatically flagged for deeper investigation. Developed using Clean Architecture principles, the system separates concerns into Domain, Application, Infrastructure, and API layers, ensuring high maintainability and extensibility. Microsoft SQL Server serves as the primary data store. The Angular frontend, styled with Tailwind CSS, provides a responsive and user-friendly interface.

Overall, the Casualty & Disaster Insurance Management System enhances operational efficiency, reduces fraud risk, improves disaster validation accuracy, and ensures transparent claim management, making it a robust and scalable solution for modern insurance organizations.